



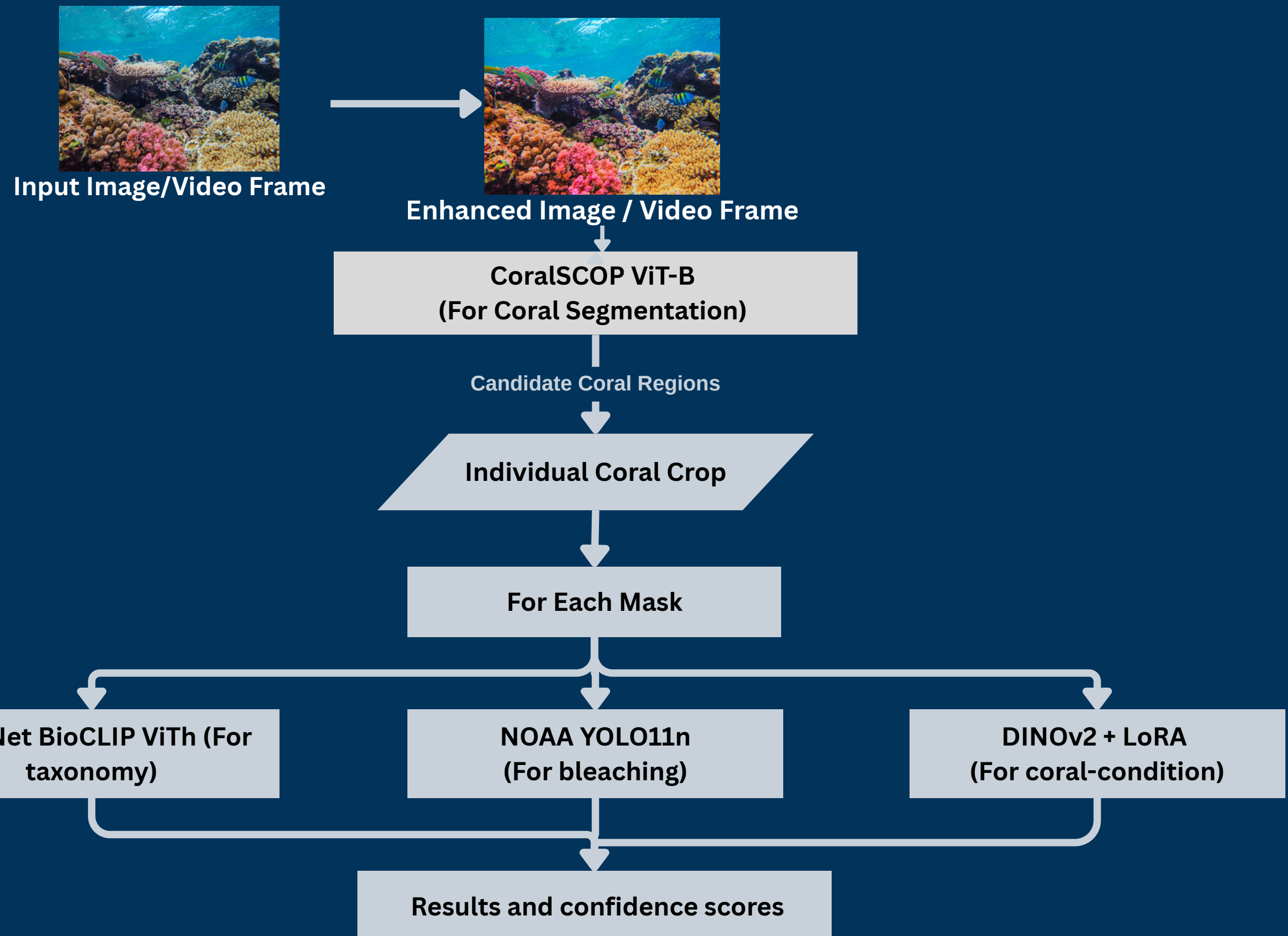
AI-BASED CORAL IDENTIFICATION AND HEALTH ASSESSMENT

PIPELINE FOR CORAL SEGMENTATION, TAXONOMY, BLEACHING AND CONDITION MONITORING

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ARCHITECTURE



CORAL SEGMENTATION

[Pre-trained model available](#)

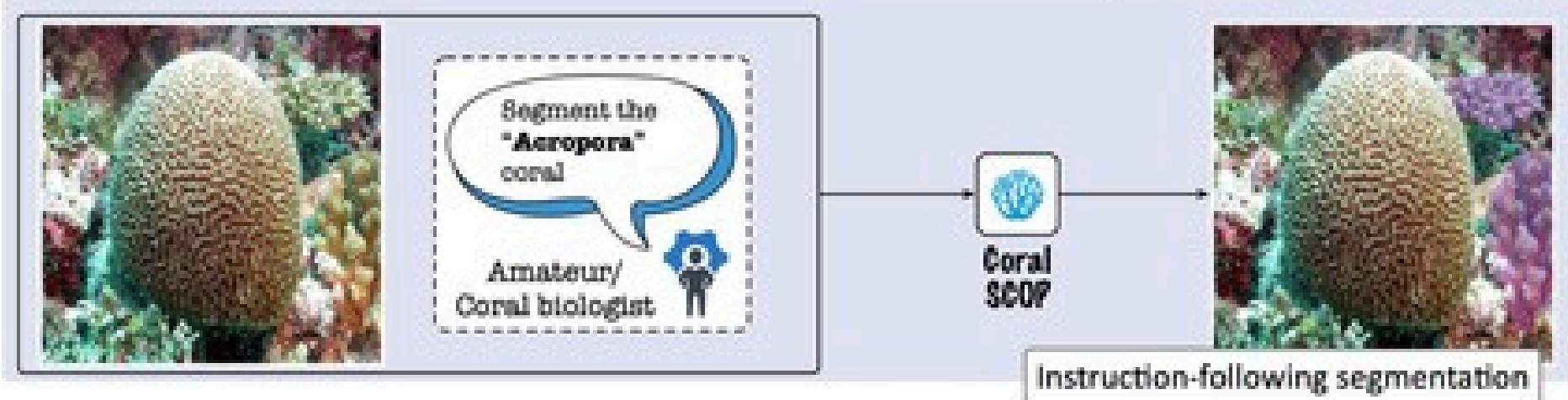
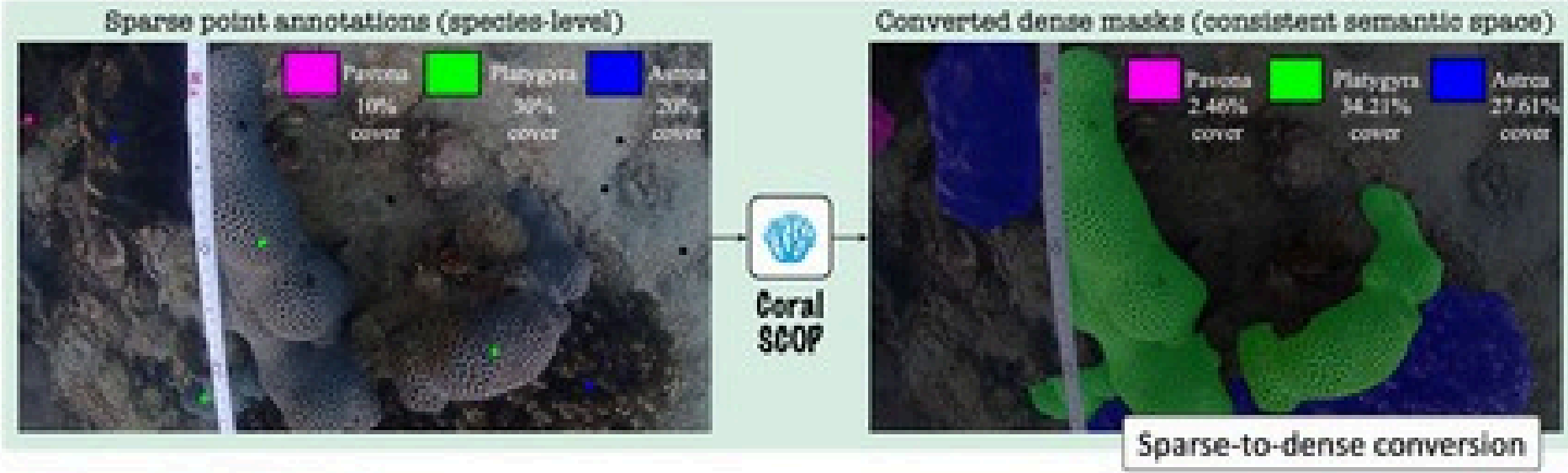
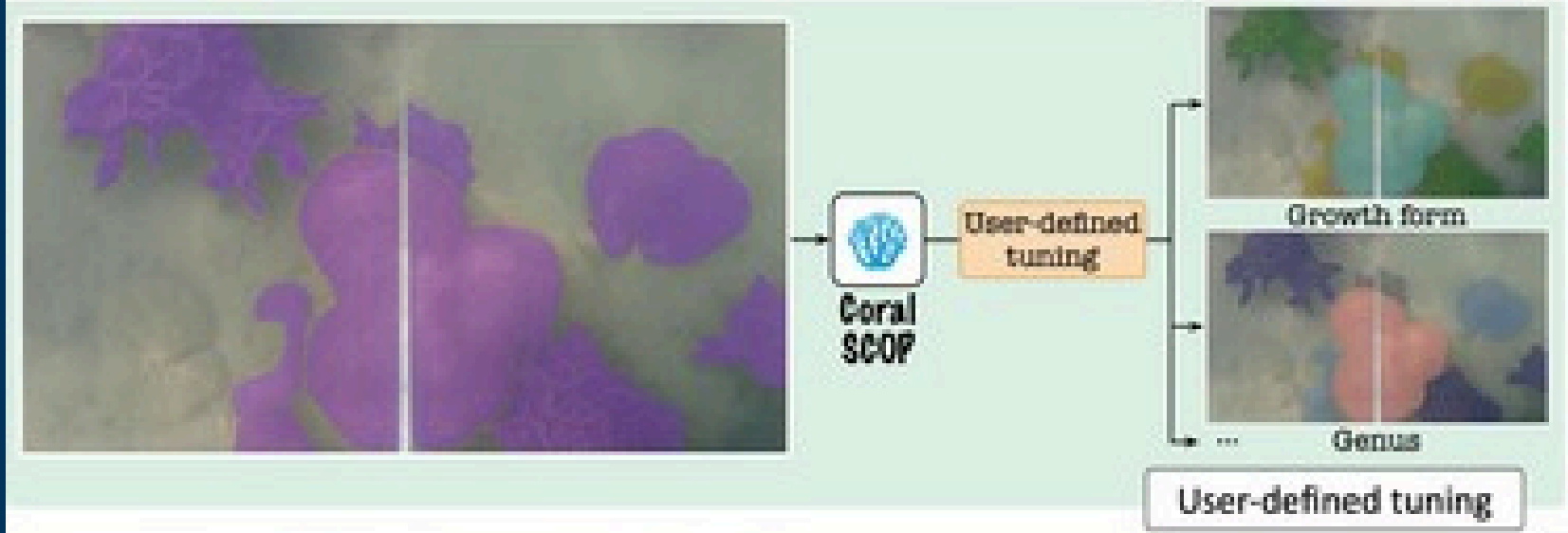
[CoralSCOP: Segment Any Coral Image on this Planet](#)

Why CoralSCOP?:

CoralSCOP is the foundation model for coral reef segmentation. It builds upon [Segment Anything Model \(SAM\)](#) but addresses the semantic-agnostic limitations and includes additional training data & strategies for underwater coral scenes. This powerful model will be made available through Reef Support's MariMap application, enhancing our ability to analyze coral reefs worldwide.

Purpose	Coral reef image segmentation using a SAM-based foundation model.
Dataset size	Trained with 41,297 images with 330,144 manually annotated coral masks.
Segmentation metrics	IoU: 59.02% automatic, 62.90% with 1 point, 74.00% with point + box.
Model size	ViT-B checkpoint: 382 MB .
Expected inference time	200-400ms for T4
Known limitations	Performance can degrade under poor visibility, colour distortion, occlusion, complex benthic backgrounds and imagery significantly different from the training data
Segmentation type	Dense, instance-aware and semantic coral segmentation.

CoralSCOP: Segment Any Coral Image on this Planet



the inaugural foundation model proposed for dense coral segmentation. CoralSCOP could serve user-defined tuning and sparse-to-dense conversion to obtain more fine-grained, reliable and accurate coral statistics. The mask-referring segmentation and instruction-following segmentation enable both amateurs and coral biologists to generate coral masks

CORAL TAXONOMY

[Pre-trained model available](#)

[ReefNet/finetuned-bioclip](#)

Coral image classification presents significant challenges for traditional CNNs due to the dense, fine-grained visual patterns inherent to marine environments. While standard CNN architectures suffice when target coral species exhibit distinct, macro-level morphological features, fine-grained distinctions require a more robust approach. Consequently, this architecture serves as an effective foundation model for downstream fine-tuning.

This model leverages BioCLIP (a foundation model pre-trained on organism taxonomy) fine-tuned on ReefNet, serving as a powerful baseline for fine-grained coral classification.

- [ReefNet database](#)
- [Methodology to finetune](#)

CORAL TAXONOMY

[ReefNet/finetuned-bioclclip](#)

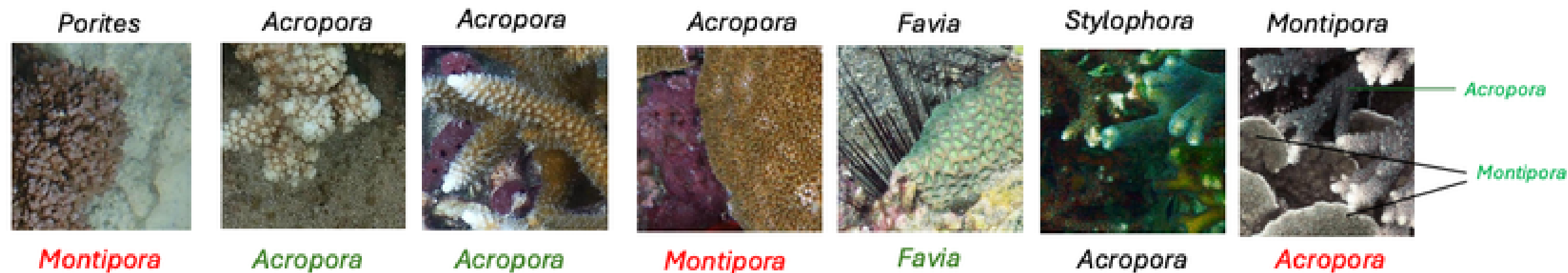
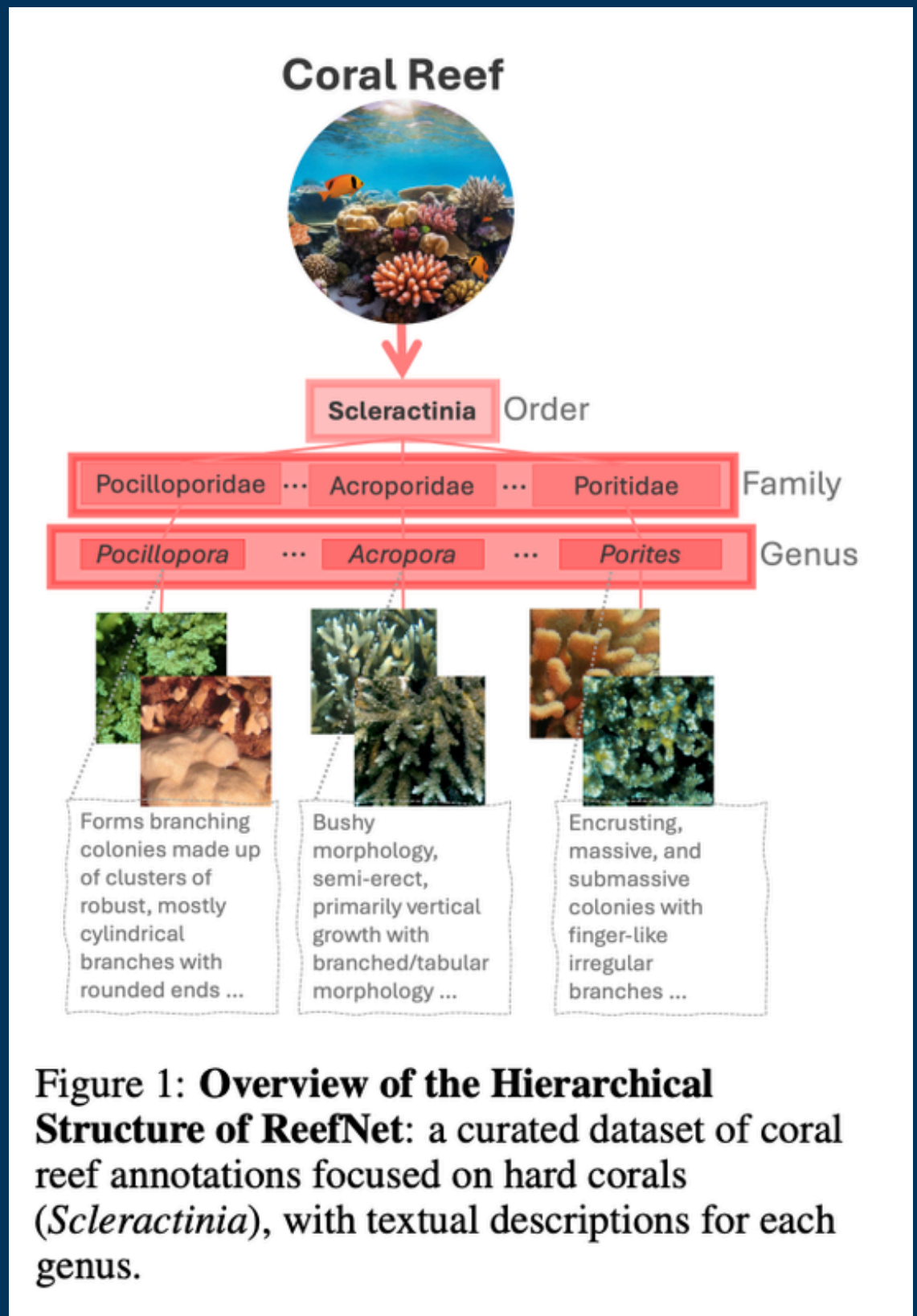


Figure 4: **Qualitative examples.** Ground truth is shown above and model prediction below; the rightmost image illustrates the challenge of assigning a single label to patches with multiple corals.



BLEACHING

[NMFS-OSI/yolo11n-cls-noaa-esd-coral-bleaching](#)

This model was trained to classify coral bleaching conditions using the YOLO11n architecture on imagery from NOAA-PIFSC Ecosystem Sciences Division (ESD) Coral Bleaching Classifier dataset. The dataset includes human-annotated points indicating healthy and bleached coral, enabling classification for marine ecosystem monitoring.

- Model Architecture: YOLO11n-cls
- Task: Image Classification
- Classes:
 - CORAL: Healthy coral
 - CORAL_BL: Bleached coral

It is trained specifically for coral bleaching:

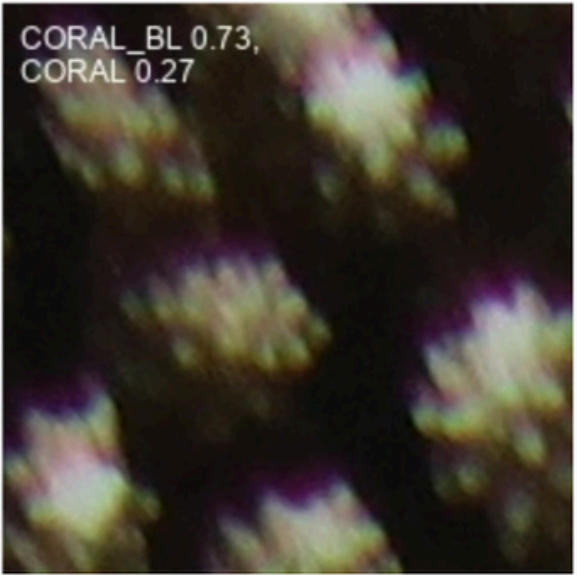
- CORAL = healthy
- CORAL_BL = bleached
- YOLO11n-cls
- 89.8% top-1 accuracy
- .pt, ONNX, NCNN and TorchScript weights available.

NMFS-OSI/yolo11n-cls-noaa-esd-coral-bleaching

Ground Truth vs Predictions



Model Inference



CORAL CONDITION AND STRESSOR ASSESSMENT

[DINOv2 + LoRA coral-condition weights](#) | [Paper](#)

It contains:

- 42,105 coral image patches
- expert labels
- training code
- DINOv2 + LoRA implementation
- the exact 8-label condition/stressor taxonomy one coral region may receive multiple labels
 - Healthy
 - Compromised
 - Dead
 - Rubble
 - Competition
 - Disease
 - Predation
 - Physical issues

Multi-label classification for multi-temporal, multi-spatial coral reef condition monitoring using vision foundation model with adapter learning”, Marine Pollution Bulletin, 2026.

The repository provides the dataset and implementation, but a directly deployable trainedcheckpoint has not been identified

DINOv2 + LoRA coral-condition weights | Paper

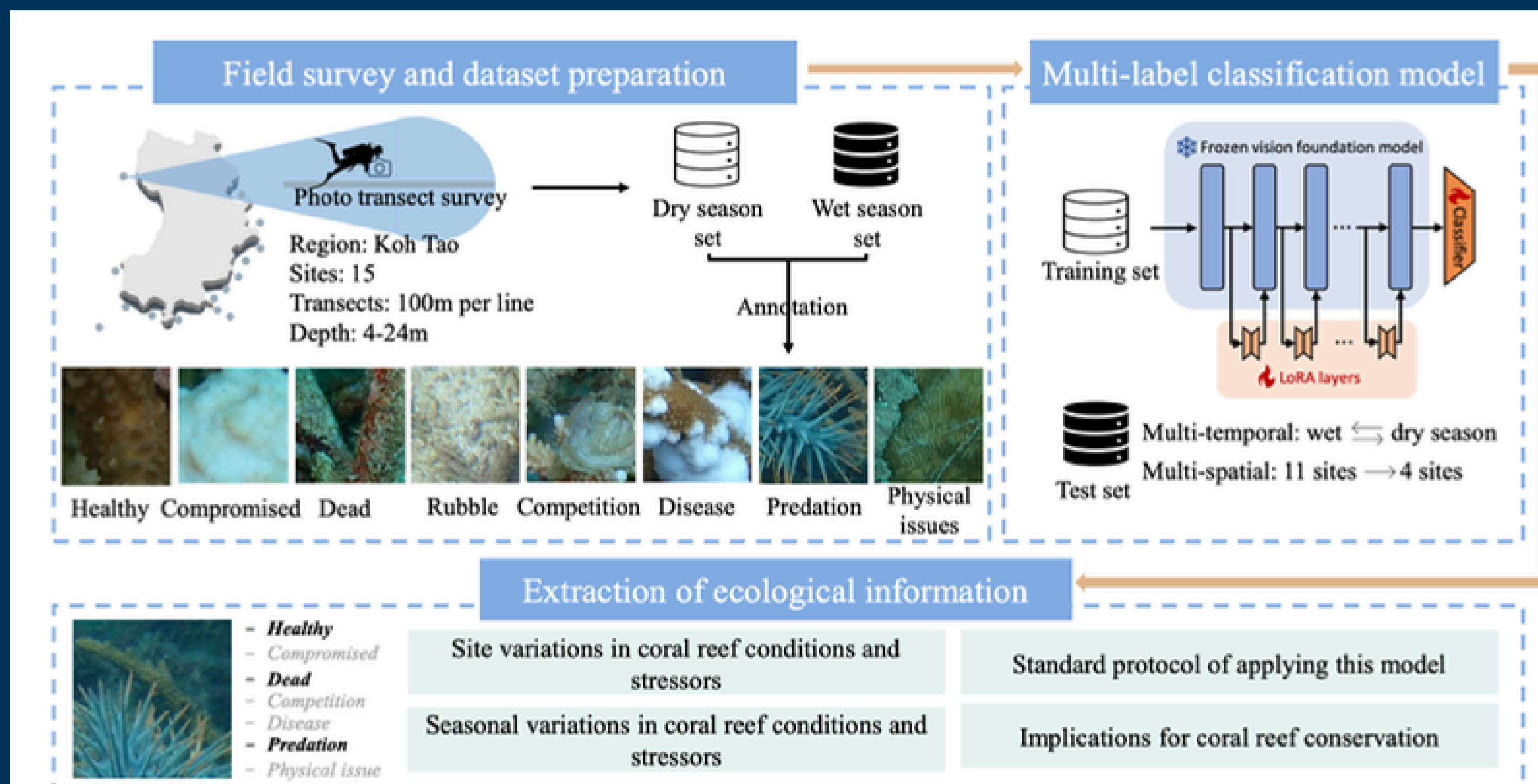


Figure 1: Flowchart illustrating the process of establishing the proposed method in this study.

Models

Function	Model	Model availability	Output	Current readiness	Licence
Segmentation	CoralSCOP ViT-B	Weights available: vit_b_coralscop.pth, 382 MB on HF	Dense coral masks; coral/non-coral, individual boundaries; semantic coral labels	High — directly usable pretrained coral model	Apache-2.0
Taxonomy	ReefNet / BioCLIP ViT-B/16	602 MB checkpoint available; official ReefNet HF repo	Taxonomic classification / similarity; ReefNet research targets genus-level hard-coral classification	Medium — checkpoint exists, but model-card provenance/output setup needs verification before production	MIT
Bleaching	NOAA YOLO11n-cls	PT + ONNX + NCNN + TorchScript available on HF	Healthy coral vs bleached coral	Very high — immediately runnable	AGPL-3.0
Coral condition	DINOv2-G + LoRA	Code + 42,105 labeled patches available; trained checkpoint not available	8-label multi-label condition/stressor prediction	Research-ready, will have to Pretrain the Weights	MIT repository

IMPLEMENTATION ROADMAP

